Dept. of Computer and Communication Engineering
Patuakhali Science and Technology University
6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2022
Course Code: CCE 323 Course Title: Optical Fiber Communication
Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1 a) Differentiate between single mode fiber optic cable and multimode fiber optic cable. 7

b) Describe different advantages of optical fiber communication

2 a) What are some technical specifications that are typically associated with optical cables, and how do these specifications influence their performance? 2

b) i. Describe the different types of fiber buffering. 5

ii. What are the main characteristics and differences between tight-buffered and loose-tube buffered fibers?

c) "The cable must be designed so that the strain on the fiber in the cable does not exceed 0.5%"- State whether the given statement is true or false. 1

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6th Semester (Level-3, Semester-II), Midterm Examination of B.Sc. Engg. (CSE), July-December: 2022
Course Code: CCE-321 Course Title: Computer Peripheral and Interfacing
Credit Hour: 3.0 Full Marks: 15 Duration: 60 Minutes

1 a) Illustrate an 8-bit microprocessor interfaced to a 4K RAM system using a full/partial decoded addressing technique. Also show the address map corresponding to the above mentioned organization. 3

b) Perform binary addition of $(-66)_{10}$ and $(-65)_{10}$. Now determine the carry, sign, overflow, and zero flags for this addition. 2

c) Describe the process of a device initiated I/O transfer. 3

2 a) What are the key functional blocks of an IoT ecosystem, and how does each block contribute to the overall functionality and efficiency of IoT solutions? 2

b) What are the unique challenges and potential applications of the Internet of Underground Things (IoUT) and the Internet of Underwater Things (IoUT)? How do these specialized IoT networks differ from traditional IoT systems? 2

c) How can the implementation of IoT in smart cities improve urban living conditions and what are the potential security and privacy challenges associated with widespread IoT deployment in such environments?

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Patuakhali Science and Technology University
Department of Computer Science and Information Technology
6th Semester (Level-3, Semester-II), **Midterm Examination** of **B.Sc. Engg.(CSE)**, July-Decem./2022, Session: 2019-20
Course Code: CIT-323 Course Title: Simulation and Modeling
Full Marks: 15 Duration: 1 hour

1. Define the following terminologies: i) System ii) Model iii) Simulation iv) Analytical solution. Demonstrate different ways in which a system might be studied. 03

2. What are the steps of simulation for a single-server queueing system? Describe the problem statement and intuitive explanation of a single-server queueing system. 05

3. Write down the difference between monte-carlo and stochastic simulation. 02

4. A drunkard moved from a point to a destination and takes step in four directions, backward, forward, right, and left. The probabilities associated with these are 10%, 50%, 15% and 25% respectively. The distances covered to backward, forward, right, and left are 30 cm, 80 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales. [USE Random Number 67, 23,91,95,84,56,77,76,18,82,45,55,29,68,21,15,86,00,89,43] 05

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Department of Computer Science and Information Technology
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Course Code: CIT-323 Course Title: Simulation and Modeling
Full Marks: 15 Duration: 1 hour

1. Define operating system. Describe the purposes of an operating system. Illustrate and describe the general structure of a clustered system. 03

2. We have stressed the need for an operating system to make efficient use of the computing hardware. When is it appropriate for the operating system to forsake this principle and to "waste" resources? Why is such a system not really wasteful? 04

3. What is data inconsistency? Mention the solution to the critical-section problem. 04

4. Write the limitation of mutex locks. How to solve it using semaphore? 04

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Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology(CSIT)

Mid Term Examination: July-December 2022
PART-B
Course Code: CIT 322 | Course Title: Operating Systems Sessional
Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 08

[Answer the questions]

1. Running a C/C++ Program in Linux 2

2. Use the commands to 06
   i. Show home directory of your user
   ii. know what files are in the directory you are in
   iii. go to a directory
   iv. create a folder or a directory
   v. delete files and directories
   vi. create a file
   vii. To know more about a command and how to use it
   viii. make a file executable and to change the permissions granted to it in Linux
   ix. move files through the command line
   x. locate a file in a Linux system
   xi. check your connection to a server
   xii. move some data, usually text into a file

3. Use the commands to 06
   i. Show home directory of your user
   ii. know what files are in the directory you are in
   iii. go to a directory
   iv. create a folder or a directory
   v. see the available disk space in each of the partitions in your system
   vi. create a file
   vii. move some data, usually text into a file
   viii. any command to be done with administrative or root privileges
   ix. compress files into a zip archive
   x. know the disk usage of a file in your system
   xi. check your connection to a server
   xii. create a folder or a directory

Patuakhali Science & Technology University (PSTU)
Department of Computer Science and Information Technology(CSIT)
Final Examination: July-December 2022
Course Code: CIT 322 | Course Title: Operating Systems Sessional
Session: 2019-20, Program: B.Sc. Engg.(CSE), Semester: 6th

Marks – 70

[Answer the marked questions from Set A , Set B]

SET-A

1. The following processes are being scheduled using a FCFS scheduling algorithm.[USE CONCURRENT PROCESS] 25

Process	Priority	Burst Time	Arrival Time
P1	2	15	0
P2	1	20	0
P3	4	20	20
P4	2	20	25
P5	3	5	45
P6	2	15	55

   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?
   iii.

2. The following processes are being scheduled using a preemptive, SJF scheduling algorithm.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   Show the execution of these processes using the following scheduling algorithms: **SJF**.
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?
   iii.

3. Create some process and apply round-robin CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?

4. Create some process and apply priority CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?

5. Create some process and apply First Come First Serve (FCFS) CPU scheduling algorithm on them.[USE CONCURRENT PROCESS AND DATASET FROM **Q1**] 25
   i. What is the turnaround time for each process?
   ii. What is the waiting time for each process?